UNIVERSITY OF OTAGO EXAMINATIONS 2023

COMPUTER SCIENCE

COSC343

Artificial Intelligence

Semester Two

(TIME ALLOWED: 3 HOURS)

This examination paper comprises 6 pages

Candidates should answer questions as follows:

Candidates should answer **All** questions

Marks are shown thus: (2)

The total number of marks for this exam is 60

The following material is provided:

Nil

Use of calculators:

Any model of calculator provided that is battery powered, silent, truly portable and free of communication capabilities

Candidates are permitted copies of:

No additional material

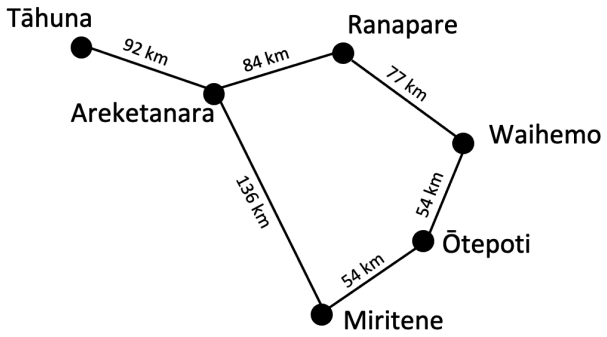
Other Instructions:

Nil

TURN OVER

1. **State-space search**

The map below shows navigable routes from Ōtepoti to other towns along with the distance of each route. The table underneath shows straight-line distances (in km) between every town.



	Ōtepoti	Miritene	Areketanara	Waihemo	Ranapere	Tāhuna
Ōtepoti	0	49	111	47	89	171
Miritene	49	0	106	90	110	157
Areketanara	111	106	0	107	58	62
Waihemo	47	90	107	0	63	167
Ranapere	89	110	58	63	0	113
Tāhuna	171	157	62	167	113	0

(a) For this set of locations, and the navigable routes between them, define a state space which can be systematically searched. Explain what a state space is and how look-ahead search works to provide an agent with a sequence of actions to navigate from one town to another. (4)

(b) In the information provided above, what constitutes the heuristic and what the transition model that enables **A* search** of the route from one town to another? (2)

(c) Carry out **A* graph search** that finds the optimal route from Ōtepoti to Areketanara and show the entire search as a tree/graph. For each node show its state, associated cost c, heuristic h, and the value of the evaluation function f. Label clearly each *explored/expanded* node with a sequence number t, starting from 0, indicating the order of node expansion (i.e. the sequence number of the root node will be $t = 0$, the node expanded after that $t = 1$ and so on; label the selected *goal* node with the last sequence number t; nodes on the fringe that never get expanded should not have a sequence label). (6)

2. **Decision trees**

Consider the following set of example data, concerning a fictitious South Island Super 12 rugby team.

H	*M*	*O*	*W*
Play_at_home	Pitch_muddy	Opposition_from_S_Africa	Win
T	T	F	T
T	F	F	F
T	F	T	T
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F
F	T	T	F
F	F	F	F

We are trying to learn a function which takes as input the boolean variables H (or Play_at_home), M (or Pitch_muddy) and O (or Opposition_from_S_Africa), and which generates as output the boolean variable W (or Win).

(a) Write down a decision tree which is consistent with these examples, starting with the split on the H attribute. (6)

(b) What result does your tree predict for the case of the team playing away on a muddy pitch against a team not from South Africa? (1)

(c) Why might a fully *consistent* tree not be the best predictor of the game result? What could be done to alleviate the problem? (2)

(d) Informally explain the principle for building an *optimal* splits decision tree. You don't have to explain the mathematics or give formulas. (3)

3. **Classification and regression**

Consider the following four hypotheses:

$\hat{y}_1 = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0,$
$\hat{y}_2 = \sigma_s(w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0),$
$\hat{y}_3 = \sigma_h(w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0),$
$\hat{y}_4 = w_1 x_1 x_2 + w_2 x_2^2 + w_3 x_1^2 x_2 + w_4 x_3,$

where $\sigma_s(v) = \frac{1}{1+e^{-v}}$, $\sigma_h(v) = \begin{cases} 1 & v > 0 \\ 0 & v \leq 0 \end{cases}$, $x_i \in \mathbb{R}$, and $w_i \in \mathbb{R}$.

(a) Identify the model/hypothesis name that each is known by (use the index of $\hat{y}_i$ to clearly identify which name refers to which hypothesis). (2)

(b) For each of those hypotheses, identify whether it is best suited for classification, regression, or for both. What loss function would be appropriate to use for training in each case? (4)

(c) Which of the above can be trained to fit a supervised dataset with the linear least squares method and which one with steepest gradient descent optimisation? (2)

(d) Assume you have a dataset with a data point $\mathbf{x} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$ with the corresponding target value $y = 0.75$. State which of the above hypotheses can be consistent on this dataset. For those that cannot be consistent, explain why. (4)

4. **Learning**

(a) What is the difference between a *model-based* (also referred to as *state-based*) and a *reflex* agent? (2)

(b) Given a powerful learning model that, after training, is perfectly *consistent* on all the training data, how can we tell if it generalises well? (2)

(c) Explain steepest gradient optimisation of the parameters of a *hypothesis* in terms of a state-space search. What constitutes the state, actions and evaluation function? What is the role of the gradient in terms of the search? Assuming the goal of the search is to find a minima, what can be said about the completeness and optimality of this search? (5)

(d) Reinforcement learning is just supervised learning with the additional complication that the agent is responsible for generating its own training data. Do you agree? Explain why or why not. (2)

(e) What does the term *epoch* refer to in machine learning? (1)

5. **Neural networks**

Consider an artificial neural network model with the output described by the following equation

$$\hat{Y} = \sigma_s\Big(\sigma_r(X \cdot W_1 + b_1) \cdot W_2 + b_2\Big),$$

where:

X is an $N \times D$ matrix of inputs,
W_1 is a $D \times U$ matrix,
W_2 is a $U \times K$ matrix,
b_1 is a U-dimensional vector,
b_2 is a K-dimensional vector,
$\hat{Y}$ is a $N \times K$ matrix,

$\sigma_s(v) = \frac{1}{1+e^{-v}}$,

$$\sigma_r(v) = \begin{cases} v & v > 0 \\ 0 & v \leq 0, \end{cases}$$

and $\cdot$ signifies the dot product.

(a) What type of neural network is this, how many layers does it have and what are the common names of the activation functions it's using? (2)

(b) Draw a diagram of this network if it was to accept 2 inputs, have 3 neurons in each of its hidden layers, and 1 output (no need to label the weights and biases, but do label inputs and outputs). (2)

(c) Suppose you are to configure this network for processing a mini-batch of 128 black and white, 8x8 pixels images from the MNIST dataset for the purpose of classifying each image into one of 10 possible classes. Choose appropriate values of N, D, U, and K. Explain in each case whether your choice is dictated by the problem/data or whether it's an arbitrary choice. (4)

(d) Can the perceptron learning rule be used to train this network? Explain why or why not. (3)

(e) Under what condition is this model a *universal classifier*? (1)

END